Formulas of the InferMesh Gateway

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Abstract

Prefix KV is reused only if the next request is assigned to a replica that still holds those blocks. That assignment is SGLang's cache-aware predicates (match > 0.3, imbalance 64 and 1.5). A miss is GREEDY$(d{=}2)$ of Azar *et al.* Retries are admitted from an SRE 10% credit ledger. A consecutive-5xx worker is ejected for $T = T_0 k$ (Envoy), $T_0 = 30\,\text{s}$, cap $300\,\text{s}$. Retry-After is RFC 9110. Three unpublished extras are marked *assumed*: P2C on miss, ETIF weight $\alpha = 0.6$, DistServe C_i in replay only. Defaults are those of the running code.

Index Terms

LLM serving, prefix cache, power of two choices, retry budget, outlier ejection

I. Introduction

Let p be a prompt and W the set of replicas that still accept work. A transformer reuses KV for a prefix of p only on a replica that holds those blocks [9]. Uniform random assignment ignores the cache; sticky assignment pins $L(w)$. Independently, unbounded retries after 5xx or 429 amplify a short outage into Bronson metastability [8].

InferMesh maps each request through four published special cases: SGLang cache-aware selection [9]; GREEDY(2) [1], [2]; SRE's 10% retry ratio [4]; Envoy $T = T_0 k$ [5]. Cool-down headers follow RFC 9110 [6]. Replay service times use the DistServe Appendix A *form* [7] with unpublished C_i.

A line marked *assumed* is executed and has no published derivation. Simulator-only identities are in Section XI. Defaults are the shipped values.

Notation. Pool W for a model: the breaker admits the worker and waiters $< Q$. Load $L(w)$ is requests in flight. Prompt string p. Match depth d. Attempts $A = 2$.

II. Prefix

$$p = \mathsf{join}(\mathsf{role} : \mathsf{content})$$

over chat messages, newline-separated. Completions use the raw prompt string. Embeddings skip the trie.

III. Token reserve (ETIF)

Heuristic, not a tokenizer. Let c be the sum of content lengths in the messages (roles are not counted), and let N_{out} be the client's `max_tokens` (or 256 if omitted).

$$\hat{n} = \max(1, \lfloor c/4 \rfloor) + N_{\text{out}}.$$

Drop N_{out} when `etif_count_output_tokens` is false. Reserved until the response ends. c is not $|p|$: the trie string includes the role labels.

IV. Selection

Problem. Send p to a worker that already holds a long prefix of p, unless that pins load.

SGLang cache-aware uses CLI defaults cache-threshold 0.3, balance-abs-threshold 64, balance-rel-threshold 1.5; load is in-flight count. Imbalanced iff both

$$\max L - \min L > 64 \quad \text{and} \quad \max L > 1.5 \min L.$$

Otherwise a hit iff there exist live tenants of the longest match with

$$d \geq d_{\text{min}} \quad \text{and} \quad d > 0.3|p|,$$

where d_{min} is the configured character floor. Then pick $\arg\min L$ among those tenants.

Here: the same predicates. On miss *or* imbalance, power-of-two choices, not SGLang's min-load (random tie-break). Match rate in SGLang is $d/|p|$ in characters; same here. Insert of p on dispatch, as SGLang inserts at selection.

V. Power of two choices

Theorem 1 (Azar–Broder–Karlin–Upfal [1]): n balls, n bins, each ball into the least loaded of $d \geq 2$ uniform bins: max load $(\log\log n)/(\log d) + O(1)$ w.h.p. For $d = 1$ it is $\Theta(\log n/\log\log n)$.

Theorem 2 (Supermarket [2]): Poisson arrivals rate λn, $\lambda < 1$, exponential service; $d = 2$ cuts expected wait exponentially versus $d = 1$.

Here: $d = 2$. Sample without replacement, proportional to worker weight. Compare scores, not raw queue length. The theorems apply only while the score is a live queue; they do not cover a stale or mixed score.

VI. Score

Let $T(w)$ be tokens in flight, $U(w) \in [0, 1]$ engine KV util, $q(w)$ engine wait depth. Write $e = \min(T/S, 1)$ and $q' = \min(q/q_0, 1)$. Then

$$s(w) = \begin{cases} e & \text{openai or stale } U, \\ \alpha e + \beta q' + (1 - \alpha - \beta)U & \text{otherwise.} \end{cases} \tag{1}$$

Defaults: $\alpha = 0.6$, $\beta = 0$, $q_0 = 8$, $S = 20 \cdot 1024$. Stale means older than 3 poll intervals. $\alpha = 0.6$ is *assumed* (no published mix). With $\beta = 0$ the vLLM term is $0.4U$.

VII. Trie

Character radix (Morrison PATRICIA [3]): one child per first character, split on LCP. Tenants stamped on every node of an insert. Match may stop inside an edge.

TTL 300 s per attribution. Cap 2^{26} characters per worker. Over cap: drop LRU leaves until 0.9 of the cap (*assumed* headroom; SGLang uses a periodic node LRU). LCP length: binary search on slices, $O(\log n)$ Python steps, $O(n)$ bytes compared.

VIII. Retry budget

SRE [4]: retry while the retry ratio stays below 10 percent; amplification at most about 1.1 versus about 3 with a per-request cap of three. Envoy RetryBudget: default 20 percent of active+pending, floor concurrency 3.

Here: credit ledger, mode static,

$$c \leftarrow \min(c_{\max},\, c + 0.10), \tag{2}$$

$$\rho = \rho_0(1 - \alpha_\rho h), \quad \rho_0 = 1,\ \alpha_\rho = 0. \tag{3}$$

Admit a retry iff $c \geq \rho$; then $c \leftarrow c - \rho$. With $\alpha_\rho = 0$, $\rho = 1$: one retry per ten requests in the mean, burst $c_0 = 3$, cap $c_{\max} = 10$. $h = d_w/|p|$ for the fallback worker; unused while $\alpha_\rho = 0$.

Mode headroom (*assumed*, off): deposit $\max(0, \min(R/(\rho_0\lambda) - 1, 1))$ once λ is known over a window of at least 1 s. Local reroutes (open breaker, full queue) skip the ledger. Hard cap $A = 2$ attempts, distinct workers. Retry only before the first byte, on 429, 5xx, or transport. Timeouts are not retried.

IX. Circuit breaker

Envoy outlier [5]: five consecutive 5xx. Real ejection time equals base ejection time times consecutive ejections, capped by max ejection time. Defaults 30 s and 300 s.

$$T(k) = \min(T_0 k,\, T_{\max}), \qquad T_0 = 30,\ T_{\max} = 300.$$

First open $k = 1$ implies $T = 30$ s. Success sets $k = 0$. Explicit Retry-After cool-down does not increment k.

States: CLOSED $\rightarrow$ OPEN after 5 consecutive 5xx (not 429) $\rightarrow$ HALF-OPEN after T, one probe. 429 opens for the header delay, unmultiplied.

Not taken from Envoy: max ejection percent (default 10 percent); the interval that decrements the multiplier while healthy (here a success zeros k).

X. Cooldown and queue

RFC 9110 §10.2.3 [6]: Retry-After is an HTTP-date or delay-seconds. Invalid means ignore. Else the largest vendor reset (reset headers, Anthropic RFC 3339 / epoch). Cap 300 s. Never ∞.

Semaphore $C = 20$ in-flight per worker. Waiters $\leq Q = 40$, wait ≤ 30 s, else 503 with Retry-After. Routable while waiters $< Q$.

XI. Replay benches (not the gateway)

Prefill time, DistServe [7] Appendix A *form* $T = C_1(...) + C_2(...) + C_3$, constants fitted per model/GPU and unpublished:

$$t_{\text{prefill}}(u) = c_1 u + c_2 u^2 + c_3$$

with $c_1 = 4 \cdot 10^{-5}$, $c_2 = 3 \cdot 10^{-9}$, $c_3 = 0.02$, ITL $0.02\,\text{s}$ *assumed*. Claim policy order, not seconds.

SLO in the replay metrics: TTFT $0.25\,\text{s}$, TPOT $0.10\,\text{s}$ (DistServe Table 1, OPT-13B chatbot). Benchmarks report a $2\,\text{s}$ TTFT cut on the long Mooncake prompts. Retry sim: Bronson *et al.* [8] §2.1, capacity 300 QPS, offered 280 QPS.

XII. Conclusion

Hit and imbalance are SGLang's CLI predicates. Sampling is GREEDY(2). The retry ledger is SRE 10% with $\rho = 1$. Ejection is Envoy $T = T_0 k$, $T_0 = 30\,\text{s}$, cap $300\,\text{s}$. Cool-down is RFC 9110.

Assumed, and left unfitted: P2C on miss (SGLang min-load), $\alpha = 0.6$, DistServe C_i in replay. α_ρ and headroom mode remain off.

References

[1] Y. Azar, A. Z. Broder, A. R. Karlin, and E. Upfal, "Balanced allocations," *SIAM J. Comput.*, vol. 29, no. 1, pp. 180–200, 1999.
[2] M. Mitzenmacher, "The power of two choices in randomized load balancing," *IEEE Trans. Parallel Distrib. Syst.*, vol. 12, no. 10, pp. 1094–1104, 2001.
[3] D. R. Morrison, "PATRICIA—practical algorithm to retrieve information coded in alphanumeric," *J. ACM*, vol. 15, no. 4, pp. 514–534, 1968.
[4] B. Beyer, C. Jones, J. Petoff, and N. R. Murphy, Eds., *Site Reliability Engineering*. Sebastopol, CA: O'Reilly, 2016, ch. 21.
[5] Envoy Proxy, "Outlier detection," https://www.envoyproxy.io/docs/envoy/latest/intro/arch_overview/upstream/outlier.
[6] R. Fielding, M. Nottingham, and J. Reschke, Eds., "HTTP semantics," RFC 9110, Jun. 2022, §10.2.3.
[7] Y. Zhong *et al.*, "DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving," in *Proc. OSDI*, 2024, arXiv:2401.09670.
[8] N. Bronson, A. Aghayev, A. Charapko, and T. Zhu, "Metastable failures in distributed systems," in *Proc. HotOS*, 2021.
[9] SGLang Project, "Cache-aware router," sgl-model-gateway, CLI defaults cache-threshold 0.3, balance-abs-threshold 64, balance-rel-threshold 1.5.